

TrES-2b

R-0152

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_180711 · created 2026-07-16T09:54:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458310.85719 +/- 0.00059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.043
Transit depth (Rp/Rs)^2	1.83 +/- 1.16 [%]
Orbital Inclination (inc)	82.02 +/- 2.82
Airmass coefficient 1 (a1)	0.84 +/- 0.19
Airmass coefficient 2 (a2)	0.17 +/- 0.42
Scatter in the residuals of the lightcurve fit is	22.58 %
Best Comparison Star	#2 - [452, 426]
Optimal Method	PSF photometry
Transit Duration (day)	0.039 +/- 0.04

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.135 $\pm$ 0.043 vs 0.1254 (deviation 0.21 σ)
Duration fit vs published	0.039 d vs 0.0746 d (deviation 0.83 σ)
Mid-time O–C	1.3 min
Depth SNR	2.9
Residual scatter	22.58 %

Light curve

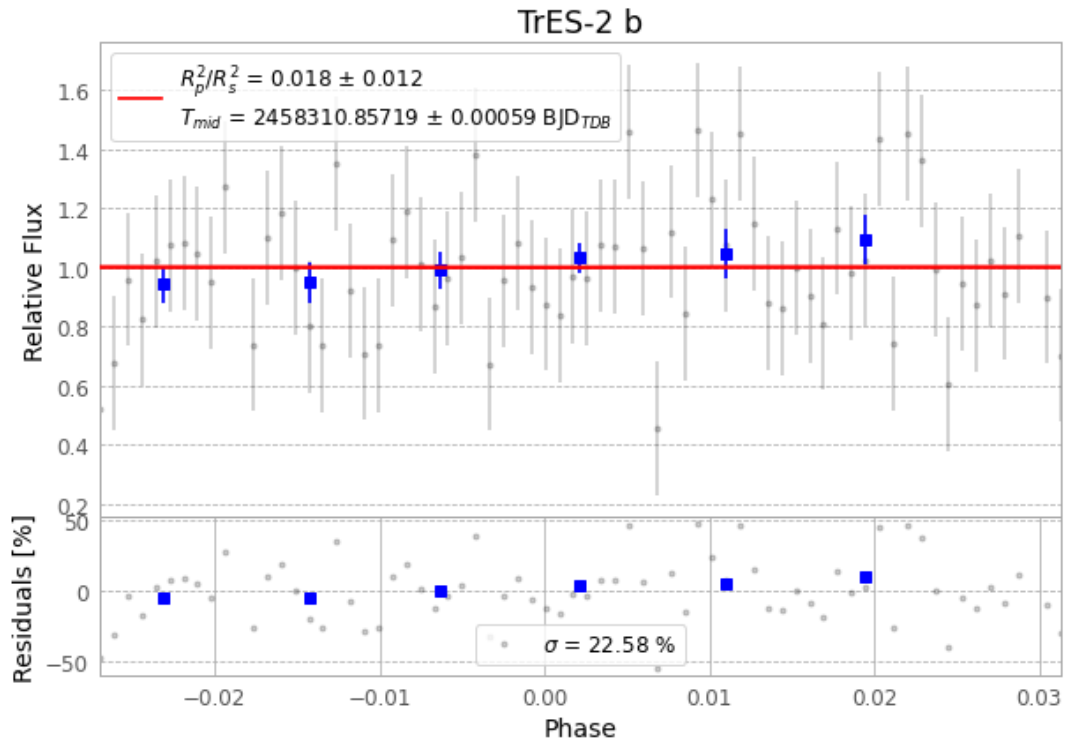


Figure 1 — FinalLightCurve_TrES-2 b_2018-07-10.png (SHA-256 4e3c395e1b4b051b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 640c1f52843140b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

70 FITS frames (46 MB), TRES-21807111065412.FITS → TRES-2180711102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180711.log (7bd28d918eff...), FinalLightCurve_TrES-2 b_2018-07-10.png (4e3c395e1b4b...), FinalLightCurve_TrES-2 b_2018-07-10.csv (98609a047cb2...)

Compute resources

Wall time 175.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 09:54Z.